



EXAM CODE: 02

Student's Full Name: Student ID:

Date: Room:

Part 1: Multiple choice (40 points)

Answer Sheet:

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

Question 1: Which classification of IP address does 225.34.5.4 belong to?

- A. Class A
B. Class B
C. Class C
D. Class D

Question 2: Which of the following is an advantage to using Port Address Translation?

- A. Lower levels of jitter
B. Lower levels of packet loss
C. Flexibility of Internet connections
D. Lower memory usage than other NAT types

Question 3: Which command would you configure on the private network interface for NAT?

- A. Router(config-if)#ip nat outside
B. Router(config)#ip nat inside gi0/0
C. Router(config-if)#ip nat private
D. Router(config-if)#ip nat inside

Question 4: Which command will allow you to verify the IP addresses assigned to a router's interface?

- A. Router#show ip dhcp bindings
B. Router#show ip interface
C. Router#show ip lease
D. Router#show ip dhcp lease

Question 5: EIU IT Department is worried about the threat of malware on the network. They want to install software that will detect worms and Trojan horses on every workstation. Which type of software should they install?

C. Software firewalls

D. Spyware

Question 6: Which cloud service is likely to be used for software development?

A. SaaS

B. IaaS

C. PaaS

D. DRaaS

Question 7: Which flags are used during the three-way-handshake process for TCP?

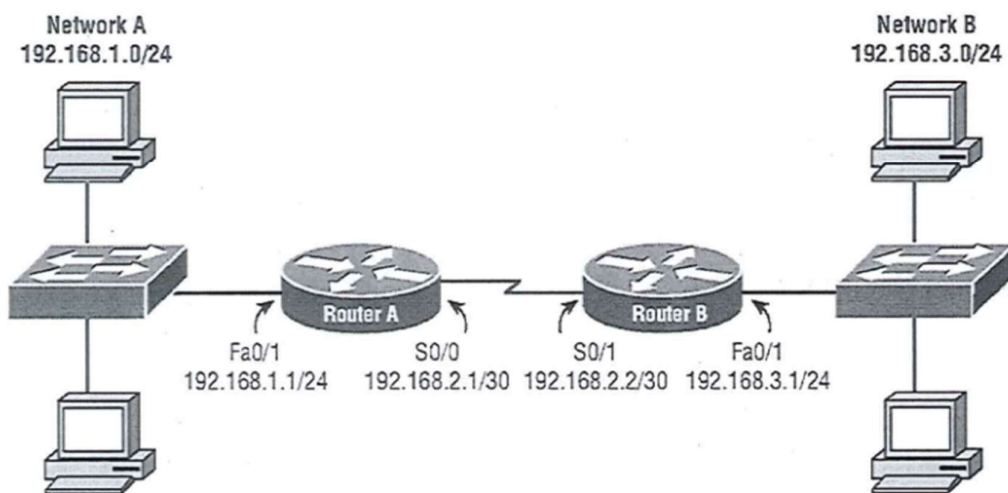
A. FIN and ACK

B. SYN and ACK

C. SYN and FIN

D. SYN and RDY

Question 8: You configure Router A in the following exhibit with a route statement to get to Network B. However, you cannot ping a host on Network B. What is the problem?



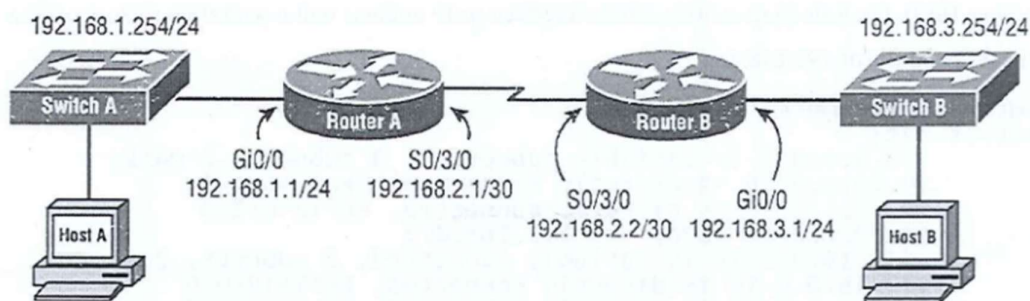
A. You must issue the ip routing command.

B. The host on Network B is misconfigured.

C. The host on Network A is misconfigured.

D. You must enter a network route on Router B for Network A.

Question 9: In the following exhibit, Host A suddenly cannot communicate with Host B. Using the ping command, which device will you ping first to diagnose the problem?



A. Switch A

B. Router A

C. Router B

D. Switch B

Question 10: Which type of NAT is used for one-to-one mapping between local and global addresses?

A. Dynamic NAT

B. Static NAT

C. NAT Overloading

D. Symmetric NAT

Question 11: Which benefit does a switch provide to a LAN?

A. Breaks up broadcast domains

B. Allows for a fast uplink port

C. Forces full-duplex on all ports

D. Breaks up collision domains

Question 12: Which protocol is connectionless and contains no flow control?

A. IP

B. TCP

C. UDP

D. ICMP

Question 13: Which network does the IP address 192.168.4.38/27 belong to?

A. 192.168.4.8/27 network

B. 192.168.4.16/27 network

C. 192.168.4.32/27 network

D. 192.168.4.64/27 network

Question 14: Which command will show the number of entries in a MAC address table?

A. Switch#show mac address-table

B. Switch#show cam count

C. Switch#show mac count

D. Switch#show mac address-table count

Question 15: You have a switch with several hundred interfaces. You only want to see the running-config for one interface, Gi3/45. Which command will allow you to see the running-config for only Gi3/45?

A. Switch#show interface gi 3/45

B. Switch#show running-config | include 3/45

C. Switch#show running-config interface gi 3/45

D. Switch#show running gi 3/45

destination address of 192.168.4.56?

```
Router#sh ip route
[output cut]
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.10.0.0/16 is directly connected, Serial0/2/0
L       10.10.1.1/32 is directly connected, Serial0/2/0
S       10.20.0.0/16 [1/0] via 192.168.4.2
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, Serial0/0/0
L       172.16.1.1/32 is directly connected, Serial0/0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, Serial0/0/1
L       192.168.1.1/32 is directly connected, Serial0/0/1
S       192.168.4.0/24 is directly connected, Serial0/0/1
S       192.168.5.0/24 [1/0] via 192.168.4.2
      198.23.24.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.23.24.0/24 is directly connected, Serial0/1/1
L       198.23.24.1/32 is directly connected, Serial0/1/1
S*      0.0.0.0/0 is directly connected, Serial0/2/0
```

- A. Interface Serial 0/0/1
- B. Interface Serial 0/0/0
- C. IP gateway of 192.168.4.1
- D. Interface Serial 0/2/0

Question 17: Which command will allow you to see the next advertisement interval for RIPv2?

- A. Router#show ip protocols
- B. Router#show ip rip database
- C. Router#show ip rip
- D. Router#show ip interface

Question 18: You need to advertise the routes 192.168.1.0/24, 192.168.2.0/24, and 192.168.3.0/24. Assuming RIPv2 is configured, which configuration will achieve this?

- A. Router(config-router)#network 192.168.0.0
- B. Router(config-router)#network 192.168.1.0
Router(config-router)#network 192.168.2.0
Router(config-router)#network 192.168.3.0
- C. Router(config-router)#network 192.168.0.0/16
- D. Router(config-router)#network 192.168.0.0 0.0.255.255

Question 19: Which is a problem with using RIPv2 in a network?

- A. Complex configuration
- B. Slow convergence
- C. The use of broadcasts
- D. Routing support for classless networks

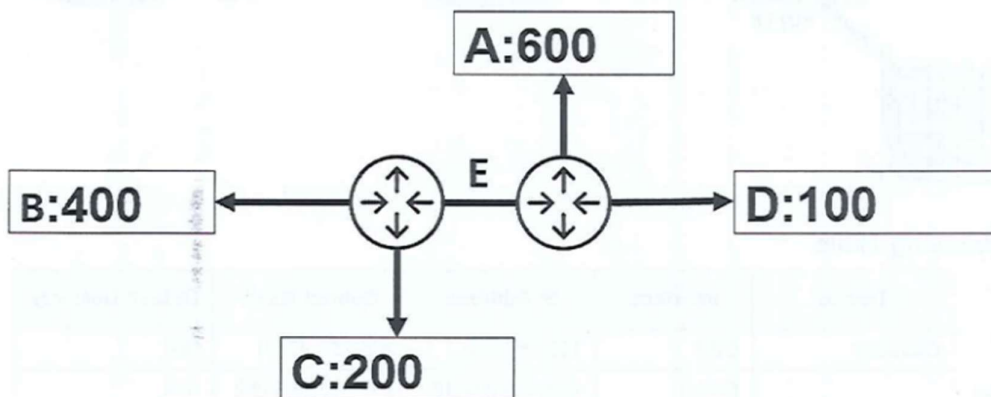
Question 20: Which command will allow you to verify the IPv6 addresses configured on a router?

- A. Router#show ipv6
- B. Router#show ip interfaces brief
- C. Router#show ipv6 interfaces brief

D. Router#show ipv6 brief

Part 2: Writing and practice (60 points)

Question 1 (20 points): Given the following topology with a network address of 172.16.96.0/20. Perform classless subnetting (VLSM) and show the results. (Network, First IP, Last IP, Broadcast)

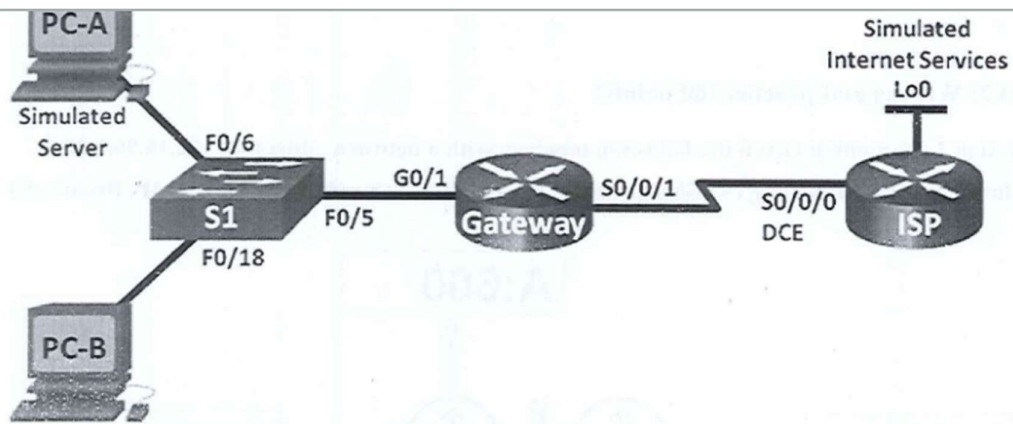


Answer:

Major Network: 172.16.96.0/20
Available IP addresses in major network: 4094
Number of IP addresses needed: 1302
Available IP addresses in allocated subnets: 1914
About 47% of available major network address space is used
About 68% of subnetted network address space is used

Subnet Name	Needed Size	Allocated Size	Address	Mask	Dec Mask	Assignable Range	Broadcast
A	600	1022	172.16.96.0	/22	255.255.252.0	172.16.96.1 - 172.16.99.254	172.16.99.255
B	400	510	172.16.100.0	/23	255.255.254.0	172.16.100.1 - 172.16.101.254	172.16.101.255
C	200	254	172.16.102.0	/24	255.255.255.0	172.16.102.1 - 172.16.102.254	172.16.102.255
D	100	126	172.16.103.0	/25	255.255.255.128	172.16.103.1 - 172.16.103.126	172.16.103.127
E	2	2	172.16.103.128	/30	255.255.255.252	172.16.103.129 - 172.16.103.130	172.16.103.131

Question 2 (40 points): In this scenario, an ISP has allocated the public IP address space of 209.165.200.224/27 to a company. This provides the company with 30 public IP addresses. The addresses, 209.165.200.225 to 209.165.200.241, are for static allocation and 209.165.200.242 to 209.165.200.254 are for dynamic allocation. A static route is used from the ISP to the gateway router, and a default route is used from the gateway to the ISP router. The ISP connection to the Internet is simulated by a loopback address on the ISP router.



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Gateway	G0/1	192.168.1.1	255.255.255.0	N/A
	S0/0/1	209.165.201.18	255.255.255.252	N/A
ISP	S0/0/0 (DCE)	209.165.201.17	255.255.255.252	N/A
	Lo0	192.31.7.1	255.255.255.255	N/A
PC-A (Simulated Server)	NIC	192.168.1.20	255.255.255.0	192.168.1.1
PC-B	NIC	192.168.1.21	255.255.255.0	192.168.1.1

Requirements:

Part 1: Build the Network and Verify Connectivity (20 points)

Part 2: Configure and Verify Static NAT (20 points)

Part 3: Configure and Verify Dynamic NAT (10 points)

----- End -----

School of CIT: Approved

Lecturer

TS. Huỳnh Tấn Phước

Dang Thai Doan



EXAM CODE: 01

Student's Full Name: Student ID:

Date: Room:

There are two parts:

Part 1: Multiple choice (40 points)

Answer Sheet:

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

Question 1: Which benefit does a switch provide to a LAN?

- A. Breaks up broadcast domains
B. Allows for a fast uplink port
C. Forces full-duplex on all ports
D. Breaks up collision domains

Question 2: Which protocol is connectionless and contains no flow control?

- A. IP
B. TCP
C. UDP
D. ICMP

Question 3: Which network does the IP address 192.168.4.38/27 belong to?

- A. 192.168.4.8/27 network
B. 192.168.4.16/27 network
C. 192.168.4.32/27 network
D. 192.168.4.64/27 network

Question 4: Which command will show the number of entries in a MAC address table?

- A. Switch#show mac address-table
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Question 5: You have a switch with several hundred interfaces. You only want to see the running-config for one interface, Gi3/45. Which command will allow you to see the running-config for only Gi3/45?

- A. Switch#show interface gi 3/45

- B. Switch#show running-config | include 3/45
- C. Switch#show running-config interface gi 3/45
- D. Switch#show running gi 3/45

Question 6: In the following exhibit, which interface or IP address will a packet be routed to for a destination address of 192.168.4.56?

```
Router#sh ip route
[output cut]
C      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
L      10.10.0.0/16 is directly connected, Serial0/2/0
L      10.10.1.1/32 is directly connected, Serial0/2/0
S      10.20.0.0/16 [1/0] via 192.168.4.2
C      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
L      172.16.0.0/16 is directly connected, Serial0/0/0
L      172.16.1.1/32 is directly connected, Serial0/0/0
C      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
L      192.168.1.0/24 is directly connected, Serial0/0/1
L      192.168.1.1/32 is directly connected, Serial0/0/1
S      192.168.4.0/24 is directly connected, Serial0/0/1
S      192.168.5.0/24 [1/0] via 192.168.4.2
C      198.23.24.0/24 is variably subnetted, 2 subnets, 2 masks
L      198.23.24.0/24 is directly connected, Serial0/1/1
L      198.23.24.1/32 is directly connected, Serial0/1/1
S*     0.0.0.0/0 is directly connected, Serial0/2/0
```

- A. Interface Serial 0/0/1
- B. Interface Serial 0/0/0
- C. IP gateway of 192.168.4.1
- D. Interface Serial 0/2/0

Question 7: Which command will allow you to see the next advertisement interval for RIPv2?

- A. Router#show ip protocols
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- A. Router(config-router)#network 192.168.0.0
- B. Router(config-router)#network 192.168.1.0
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Router(config-router)#network 192.168.3.0
- C. Router(config-router)#network 192.168.0.0/16
- D. Router(config-router)#network 192.168.0.0 0.0.255.255

Question 9: Which is a problem with using RIPv2 in a network?

- A. Complex configuration
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Question 10: Which command will allow you to verify the IPv6 addresses configured on a router?

- A. Router#show ipv6
- B. Router#show ip interfaces brief
- C. Router#show ipv6 interfaces brief
- D. Router#show ipv6 brief

Question 11: Which command will allow you to check basic connectivity at layer 3?

- A. Router#show ip route
- B. Router#ping 192.168.4.1
- C. Router#pathping 192.168.4.1
- D. Router#ip ping 192.168.4.1

Question 12: Which of the following is an advantage to using Port Address Translation?

- A. Lower levels of jitter
- B. Lower levels of packet loss
- C. Flexibility of Internet connections
- D. Lower memory usage than other NAT types

Question 13: Which command would you configure on the private network interface for NAT?

- A. Router(config-if)#ip nat outside
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- C. Router(config-if)#ip nat private
- D. Router(config-if)#ip nat inside

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- A. Malware
- B. Antivirus
- C. Software firewalls
- D. Spyware

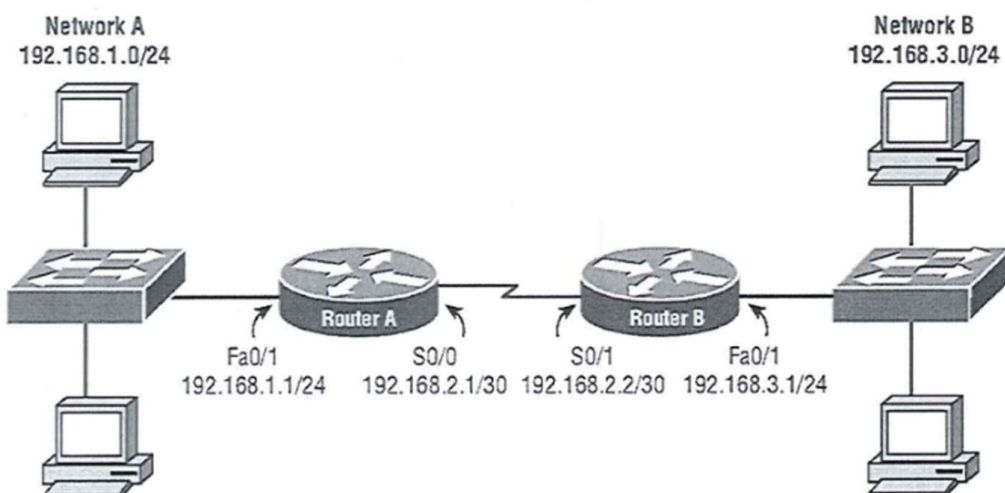
Question 16: Which cloud service is likely to be used for software development?

- A. SaaS
- B. IaaS
- C. PaaS
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Question 17: Which flags are used during the three-way-handshake process for TCP?

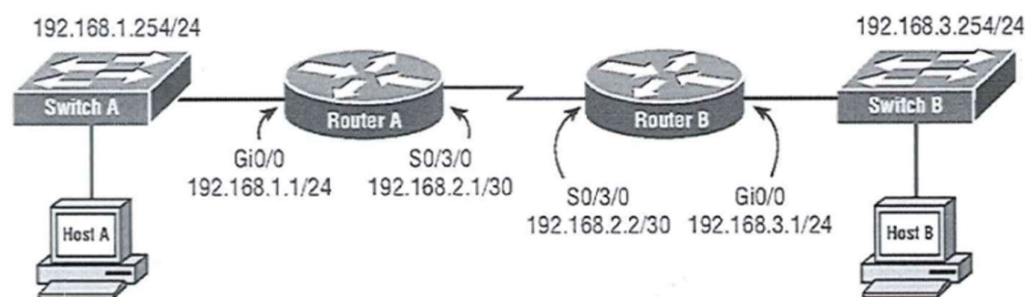
- A. FIN and ACK
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Question 18: You configure Router A in the following exhibit with a route statement to get to Network B. However, you cannot ping a host on Network B. What is the problem?



- A. You must issue the ip routing command.
- B. The host on Network B is misconfigured.
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Question 19: In the following exhibit, Host A suddenly cannot communicate with Host B. Using the ping command, which device will you ping first to diagnose the problem?



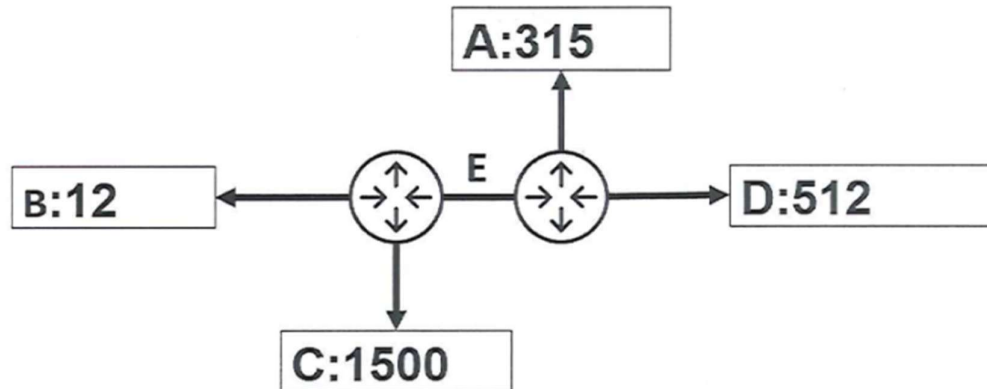
- A. Switch A
- B. Router A
- C. Router B
- D. Switch B

Question 20: Which type of NAT is used for one-to-one mapping between local and global addresses?

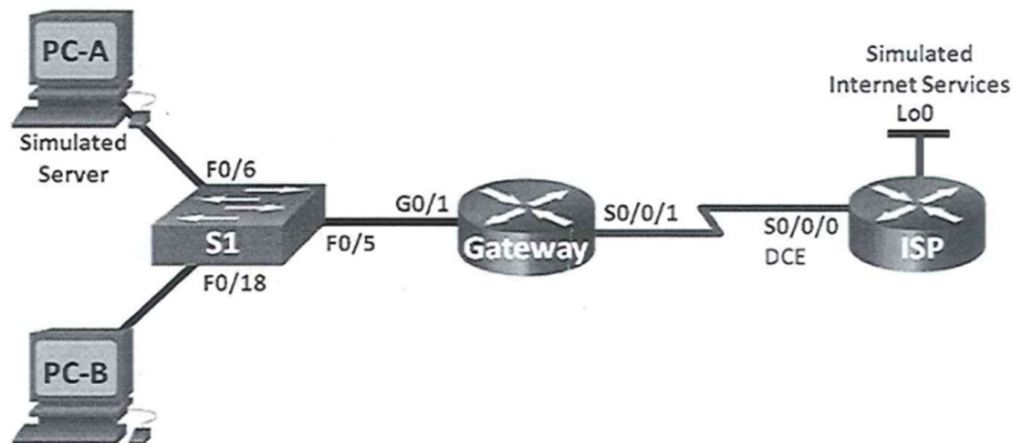
- A. Dynamic NAT
- B. Static NAT
- C. NAT Overloading
- D. Symmetric NAT

Part 2: Writing and practice (60 points)

Question 1 (10 points): Given the following topology with a network address of 172.16.0.0/16.
Perform classless subnetting (VLSM) and show the results. (Network, First IP, Last IP, Broadcast)



Question 2 (50 points): In this scenario, an ISP has allocated the public IP address space of 209.165.200.224/27 to a company. This provides the company with 30 public IP addresses. The addresses, 209.165.200.225 to 209.165.200.241, are for static allocation and 209.165.200.242 to 209.165.200.254 are for dynamic allocation. A static route is used from the ISP to the gateway router, and a default route is used from the gateway to the ISP router. The ISP connection to the Internet is simulated by a loopback address on the ISP router.



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PC-A (Simulated Server)	NIC	192.168.1.20	255.255.255.0	192.168.1.1
PC-B	NIC	192.168.1.21	255.255.255.0	192.168.1.1

Requirements:

Part 1: Build the Network and Verify Connectivity (20 points)

Part 2: Configure and Verify Static NAT (20 points)

Part 3: Configure and Verify Dynamic NAT (10 points)

----- End -----

School of CIT: Approved



TS. Huỳnh Tấn Phước

Lecturer



Dang Thai Doan



EXAM CODE: ...02..

Student's Full Name: Student ID:

Date: Room:

[Questions]

Part 1: Multiple Choices – 40 Points (20 Questions, 2 points per question)

Answer sheet:

	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

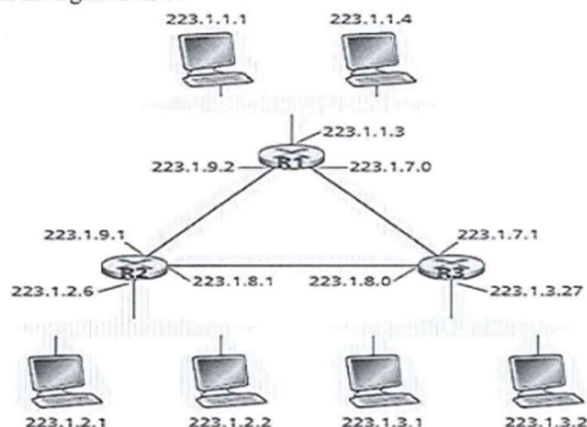
	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				

Select: X

Deselect: ⊗

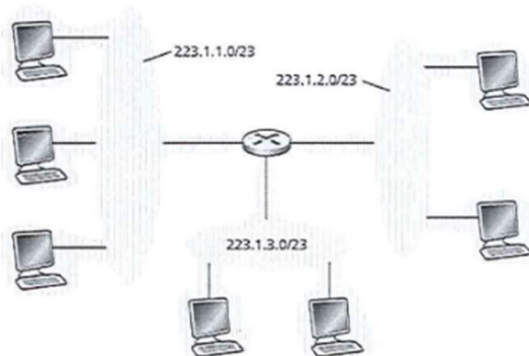
Select again: ●

1. How many subnets in the figure below?



- A. Three are 3 subnets.
 B. Three are 4 subnets.
 C. Three are 5 subnets.
 D. Three are 6 subnets.

2. Refer to the exhibit.



- A. There are 3 subnet mask address
 B. There are 3 subnet address
 B. There are 3 IP address
 D. There are 7 IP address

3. Private IP range for Class C is

- A. 0.0.0.0 to 0.0.255.255
 B. 192.168.0.0 to 192.168.0.255
 C. 192.168.0.0 to 192.168.255.255
 D. 0.0.0.0 to 0.0.0.255

4. A given network: 192.168.0.0 /24. You divide the network having 6 PCs.

Which new subnet is correct ?

- A. 192.168.0.0/26
 B. 192.168.0.0/27
 C. 192.168.0.0/28
 D. 192.168.0.0/29

5. A given network: 192.168.0.0 /24. You divide the network having 5 PCs.

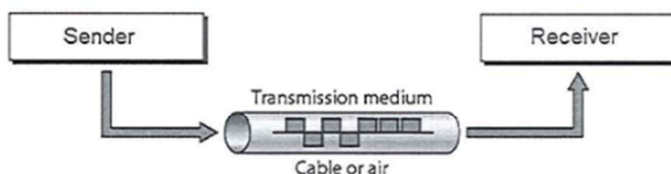
How many new subnet have you got ?

- A. 32 B. 16
C. 64 D. 128

6. Which definition is described for function of Router?

- A. A router is a networking device that is described at layer 2 in OSI, and it is used to assign IP address for local devices.
B. A router is connected to two or more data lines from different networks, and it is used to assign IP address for local devices.
C. A router is connected to two or more data lines from networks together, and it is described at layer 2 in OSI.
D. A router is connected to two or more data lines from networks together, and it is described at layer 1 in OSI.

7. The data is sent from the Sender to the Receiver as shown is happened at



- A. Transport Layer B. Session Layer
C. Data link Layer D. Physical Layer

8. The mapping information between IP address and MAC address are saved in

- A. ip table B. routing table
C. mac-address table D. arp table

9. A user is complaining that an external web page is taking longer than normal to load. The web page does eventually load on the user machine. Which tool should the technician use with administrator privileges in order to locate where the issue is in the network?

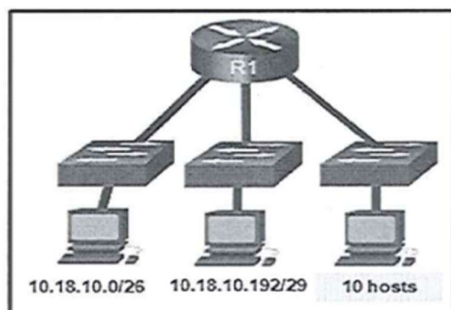
- A. ping B. nslookup
C. tracert D. ipconfig /displaydns

10. Refer to the exhibit.

Which two network addresses can be assigned to the network containing 10 hosts? Your answers should waste the fewest addresses, not reuse addresses that are already assigned, and stay within the 10.18.10.0/24 range of addresses.

- A. 10.18.10.200/28; 10.18.10.224/28 B. 10.18.10.208/28; 10.18.10.224/28
C. 10.18.10.0/28; 10.18.10.200/28 D. 10.18.10.10/26; 10.18.10.192/29

Sum



11. Which command shows the information below?

```

1 default active Fa0/1, Fa0/2, Fa0/3, Fa0/4
Fa0/5, Fa0/6, Fa0/7, Fa0/8
Fa0/9, Fa0/10, Fa0/11, Fa0/12
Fa0/13, Fa0/14, Fa0/15, Fa0/16
Fa0/17, Fa0/18, Fa0/19, Fa0/20
Fa0/21, Fa0/22, Fa0/23, Fa0/24
1002 fddi-default act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default act/unsup
1005 trnet-default act/unsup

```

- A. show ip vlan B. show routing
C. show ip interface brief D. show vlan

12. Which command shows the information below

Interface	IP-Address	OK?	Method	Status	Protoc
GigabitEthernet0/0	192.168.1.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Serial0/2/0	10.0.0.1	YES	manual	up	up
Serial0/2/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

- A. show ip route B. show routing
C. show ip interface brief D. show route

13. Cable that is used to connect Router to Router in the network is
A. Straight-through cable B. Serial cable
C. Cross-over cable D. Console cable
14. The size of an IP address in IPv6 is _____
A. 4 bytes B. 32 bits
C. 128 bits D. 100 bits
15. Which OSI Reference Layer is concerned with path determination?
A. Data link B. Session
C. Network D. Physical
16. Which of the following are examples of the Transport Layer?
A. UDP B. LLCs
C. HTTP D. Token ring
17. In computer network, NIC stands for?
A. National International Center B. Network Information Center
C. Network Interface Case D. Network Interface Card
18. What term describes a computing model where server software runs on dedicated computers?
A. client/server B. internet
C. intranet D. extranet
19. _____ has many of the characteristics of what is now being called cloud computing.
A. Internet B. Softwares
C. Hardwares D. Web Service
20. What is the network address of ip address and subnet mask:
IP: 10101100.00010000.00000010.10100000
SM: 11111111.11111111.11111111.00000000
A. 172.16.0.0 B. 172.16.2.0
C. 172.0.1.0 D. 172.16.1.0
-
-

LAN1	Subnet address	
	Broadcast address	
	First PC address	
	Last PC address	
LAN2	Subnet address	
	Broadcast address	
	First PC address	
	Last PC address	
LAN3	Subnet address	
	Broadcast address	
	First PC address	
	Last PC address	

a. Analyze and Draw a network diagram. (20 Points)

[illegible]

b. Configure routers to connect all rooms together. (20 Points)


TS. Huỳnh Tấn Phước

TS. Huỳnh Tấn Phước

EASTERN INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTING
AND INFORMATION TECHNOLOGY



EXAM CODE: 01

Final Exam – Quarter 4, Year 2022 - 2023

Course Name: Computer Network

Course Code: CSE 204

Duration: 90 minutes

Exam Type: Written test – Closed book

Student's Full Name: Student ID:

Date: Room:

There are two parts:

Part 1: Multiple choice (40 points)

Answer Sheet:

	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
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Question 1: Which command can be used on a Cisco router to display all interfaces, the IPv4 address assigned, and the current status?

- A. show ip interface brief
- B. ping
- C. show ip route
- D. show interface fa0/1

Question 2: Which statement accurately describes a characteristic of IPv4?

- A. All IPv4 addresses are assignable to hosts.
- B. IPv4 has a 32-bit address space.
- C. An IPv4 header has fewer fields than an IPv6 header has.

D. IPv4 has a 128-bit address space.

Question 3: What is the prefix length notation for the subnet mask 255.255.255.224?

- A. /25 B. /26 C. /27 D. /28

Question 4: How many valid host addresses are available on an IPv4 subnet that is configured with a /26 mask?

- A. 254 B. 190 C. 64 D. 192

Question 5: What is the most compressed representation of the IPv6 address 2001:0db8:0000:

- A. abcd:0000:0000:0000:0001?
B. 2001:0db8:abcd::1
C. 2001:db8:0:abcd::1
D. 2001:0db8:abcd::1

Question 6: Which utility uses Internet Control Messaging Protocol (ICMP)?

- A. RIP B. DNS C. ping D. NTP

Question 7: A user who is unable to connect to the file server contacts the help desk. The help desk technician asks the user to ping the IP address of the default gateway that is configured on the workstation. What is the purpose of this ping command?

- A. to obtain a dynamic IP address from the server
B. to request that the gateway forward the connection request to the file server
C. to test that the host has the capability to reach hosts on other networks
D. to resolve the domain name of the file server to its IP address

Question 8: Which protocol can be used to transfer messages from an email server to an email client?

- A. SMTP B. POP3 C. SNMP D. SMB

Question 9: Which statement is true about FTP?

- A. A client can choose whether FTP should establish one or two connections.
B. A client can download data from or upload to a server.
C. FTP is a peer-to-peer application.
D. FTP does not provide reliability during data transmission.

Question 10: A wireless host needs to request an IPv4 address. What protocol would be used to process the request?

- A. FTP B. HTTP C. DHCP D. ICMP

Question 11: A network technician suspects that a particular network connection between two Cisco switches is experiencing a duplex mismatch. Which command would the technician use to see the Layer 1 and Layer 2 details of a switch port?

- A. show interfaces
- B. show running-config
- C. show ip interface brief
- D. show mac address-table

Question 12: What is the complete range of TCP and UDP wellknown ports?

- A. 0 to 255
- B. 0 to 1023
- C. 256 to 1023
- D. 1024 to 49151

Question 13: What factor should be considered in the design of a small network when devices are being chosen?

- A. cost of devices
- B. redundancy
- C. traffic analysis
- D. ISP

Question 14: Which information is used by routers to forward a data packet toward its destination?

- A. source IP address
- B. destination data link address
- C. source data link address
- D. destination IP address

Question 15: A computer has to send a packet to a destination host in the same LAN. How will the packet be sent?

- A. The packet will be sent to the default gateway first, and then, depending on the response from the gateway, it may be sent to the destination host.
- B. The packet will be sent directly to the destination host.
- C. The packet will first be sent to the default gateway, and then from the default gateway it will be sent directly to the destination host.
- D. The packet will be sent only to the default gateway

Question 16: Which network device makes forwarding decisions based only on the destination MAC address that is contained in a frame?

- A. repeater
- B. hub
- C. Router
- D. Layer 2 switch

Question 17: IPv4 and IPv6 can coexist due to

- A. Translation
- B. Tunneling
- C. Dual Stack
- D. All of above

Question 18: Which is the right order of DHCP process?

- A. DHCPACK, DHCPRequest, DHCPOffer and DHCPDiscover
- B. DHCPOffer, DHCPDiscover, DHCPRequest and DHCPACK
- C. DHCPRequest, DHCPDiscover, DHCPOffer and DHCPACK
- D. DHCPDiscover, DHCPOffer, DHCPRequest and DHCPACK

Question 19: Which kind of virus makes a backdoor to give some information of victim?

- A. Trojan Horse
- B. Worm
- C. Virus
- D. Hacker

Question 20: What is the NAT?

- A. NAT allow public network to connect to the Internet
- B. NAT do not allow public network to connect to the Internet
- C. NAT do not allow private network to connect to the Internet
- D. NAT allow private network to connect to the Internet

Part 2: Writing and practice (60 points)

Question1. Your computer dose not connect to Internet, base on OSI model, you explain and find the way to fix it. (20 pts)

This image shows a full page of handwriting practice paper. It features multiple sets of three horizontal dashed lines, each set providing a guide for letter height and placement. The lines are evenly spaced across the entire page, leaving ample room for practicing various letters and words. There are no margins, text, or other markings on the paper.

Question2 . Divide network 192.168.0.0 /24 into 8 subnets: (10pts)

Subnet 1	Subnet mask	
	Subnet address	
	Broadcast address	
	First PC address	
	Last PC address	

Subnet 3	Subnet mask	
	Subnet address	
	Broadcast address	
	First PC address	
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Subnet 5	Subnet mask	
	Subnet address	
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Subnet 8	Subnet mask	
	Subnet address	
	Broadcast address	
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Question3 . Divide network (VLSM) 192.168.0.0 /24 into 4 subnets:

LAN1: 50 PCs

LAN2: 100 PCs

LAN3: 10 PCs

LAN4: 30 PCs

Calculate and fill in the below table (10 points)

Old subnet mask		
LAN...	Subnet mask	
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This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper appears to be from a notebook or a standard ruled sheet of paper. There is no handwriting or other markings on the page.

Lecturer

Signature

Huỳnh Tấn Phước

EASTERN INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTING
AND INFORMATION TECHNOLOGY



EXAM CODE: 02

Final Exam – Quarter 4, Year 2022 - 2023

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Course Code: CSE 204

Duration: 90 minutes

Exam Type: Written test – Closed book

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- #### D. ISP

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Question1. Your computer dose not connect to Internet, base on OSI model, you explain and find the way to fix it. (20 pts)

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Question2 . Divide network 201.10.0.0 /24 into 8 subnets: (10pts)

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	Subnet address	
	Broadcast address	
	First PC address	
	Last PC address	

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	Subnet address	
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	Subnet address	
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	First PC address	
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Question3 . Divide network (VLSM) 202.1.0.0 /24 into 4 subnets:

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Calculate and fill in the below table (10 points)

Old subnet mask		
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	Broadcast address	
	First PC address	
	Last PC address	
LAN...	Subnet mask	
	Subnet address	

Huỳnh Tấn Phước



EXAM CODE: 01

Student's Full Name: Student ID:

Date: Room:

Section A: Multiple choice questions (60 points)

There are 30 multiple choice questions. Each question is worth 02 points.

- Q1. Which devices on a network will receive an ARP request?
- A. Only the device that has the IPv4 address that the request is looking for
 - B. Only devices in the same collision domain
 - C. All devices in the layer 2 broadcast domain
 - D. All devices in the entire networks
- Q2. What is the difference between authentication and authorization?
- A. Authentication prevents unauthorized access to a network, while authorization monitors network traffic for security threats
 - B. Authentication verifies the identity of a user or device, while authorization determines what actions they are allowed to perform
 - C. Authentication encrypts data during transmission, while authorization determines network access permissions
 - D. Authentication secures wireless networks, while authorization manages network devices and configurations
- Q3. Which of the following is the purpose of the next-hop IP address in a static route configuration?
- A. It represents the interface of the local router that leads to the destination network
 - B. It determines the metric value for the route
 - C. It identifies the remote router that leads to the destination network
 - D. It specifies the IP address of the destination network
- Q4. What is the last host IP address in the subnet 10.20.24.192/27?
- A. 10.20.24.222
 - B. 10.20.24.207
 - C. 10.20.24.254
 - D. 10.20.24.206
- Q5. Which email protocol allows users to access and manage their email messages from multiple devices?
- A. SMTP
 - B. POP
 - C. HTTP
 - D. IMAP

- Q6. What is the function of a router?
- A. It connects multiple IP networks.
 - B. It controls the flow of data via the use of layer 2 addresses.
 - C. It determines the best path to send packets.
 - D. A and C
- Q7. Which of the following statements about Telnet is true?
- A. Telnet is primarily used for secure file transfers
 - B. Telnet requires additional software to establish a connection
 - C. Telnet is considered insecure and is not recommended for use
 - D. Telnet encrypts data transmitted over the network
- Q8. What is the primary function of a firewall?
- A. To encrypt data transmitted over a network
 - B. To prevent unauthorized access to a network
 - C. To monitor network traffic and detect vulnerabilities
 - D. To manage network devices and configurations
- Q9. What type of attack involves an attacker intercepting communication between two parties?
- A. Denial of service
 - B. Man-in-the-middle
 - C. Spoofing
 - D. Phishing
- Q10. Which prefix length is used for the subnet mask 255.255.255.224?
- A. /27
 - B. /24
 - C. /25
 - D. /26
- Q11. Which of the following is NOT a characteristic of Layer 2 switches?
- A. Operate at the Data Link Layer of the OSI model
 - B. Use MAC addresses for forwarding decisions
 - C. Perform routing functions between networks
 - D. Support VLAN segmentation
- Q12. What is the primary defense against unauthorized remote access to network devices?
- A. Configuring a VTY password
 - B. Configuring a default gateway
 - C. Configuring an IP address
 - D. Configuring a console password
- Q13. Why is the DHCPREQUEST message sent as a broadcast during the DHCP process?
- A. For hosts on other subnets to receive the information
 - B. To notify other DHCP servers on the subnet that the IP address was leased
 - C. For routers to fill their routing tables with this new information
 - D. To notify other hosts not to request the same IP address
- Q14. Which of the following protocols is used for receiving email?
- A. POP3
 - B. SMTP
 - C. IMAP
 - D. A and C
- Q15. What is the purpose of an Intrusion Detection System (IDS)?
- A. To prevent unauthorized access to a network
 - B. To encrypt and secure network communications
 - C. To provide redundancy and fault tolerance in a network
 - D. To monitor and detect suspicious activity on a network

- Q16.** Which command is used to save the current running configuration on a Cisco switch?
- A. write
 - B. copy startup-config running-config
 - C. copy running-config startup-config
 - D. A and C
- Q17.** What is the subnet mask of the network 10.16.32.0/20?
- A. 255.255.254.0
 - B. 255.255.240.0
 - C. 255.255.248.0
 - D. 255.255.252.0
- Q18.** When configuring a host, the gateway was not configured. How will this affect communication to and from this device?
- A. Host communication will not be affected.
 - B. No communication will be successful to or from the host.
 - C. The host will not be able to communicate with devices by name.
 - D. Communication will be limited to hosts in the local network.
- Q19.** Which of the following statements best describes ICMP protocol?
- A. It is an internet layer protocol used for error reporting and diagnostic purposes.
 - B. It is a transport layer protocol used for reliable data transmission.
 - C. It is a network layer protocol used for routing packets.
 - D. It is an application layer protocol used for remote network management.
- Q20.** What is the network address for the IP address 192.168.10.155/27?
- A. 192.168.10.160
 - B. 192.168.10.32
 - C. 192.168.10.128
 - D. 192.168.10.64
- Q21.** Which of the following IPv4 addresses would not be used in a private network?
- A. 172.32.10.10
 - B. 172.20.1.12
 - C. 192.168.100.10
 - D. A and C
- Q22.** Which TCP mechanism is used to handle out-of-order packets?
- A. Sliding window
 - B. Congestion control
 - C. Three-way handshake
 - D. Selective Acknowledgment (SACK)
- Q23.** What type of message does a device send when it needs to resolve an IP address to a MAC address?
- A. ARP request message
 - B. ARP response message
 - C. UDP message
 - D. ICMP message
- Q24.** What is the purpose of TCP's three-way handshake?
- A. To exchange acknowledgments between the sender and receiver
 - B. To terminate an established connection
 - C. To ensure data integrity during transmission
 - D. To establish a connection between the client and server
- Q25.** Which DHCP message will a client send to accept an IP address that is offered by a DHCP server?
- A. DHCPACK
 - B. DHCPREQUEST
 - C. DHCPDISCOVER
 - D. DHCPOFFER

- Q26. What is the purpose of a MAC address table in a Layer 2 switch?
- To determine the network segment of a device
 - To identify the default gateway for a network
 - To store information about connected devices' MAC addresses
 - To map IP addresses to MAC addresses
- Q27. What is the broadcast IP address in the subnet 10.20.24.0/21?
- 10.20.25.255
 - 10.20.27.255
 - 10.20.31.255
 - 10.20.24.255
- Q28. What happens if a UDP packet is lost during transmission?
- The sender automatically retransmits the lost packet
 - The sender and receiver negotiate a new connection and start the transmission again
 - The receiver sends a request to retransmit the lost packet
 - The receiver discards the lost packet and continues with the next packet
- Q29. A network technician borrows 6 bits from the 10.0.0.0/16 network to create new subnets. How many host IP addresses are there in each created subnet?
- 2046
 - 1022
 - 30
 - 62
- Q30. Which command is used to configure SSH access on a Cisco router?
- crypto key generate rsa
 - ip ssh version 2
 - ssh version 2
 - aaa new-model

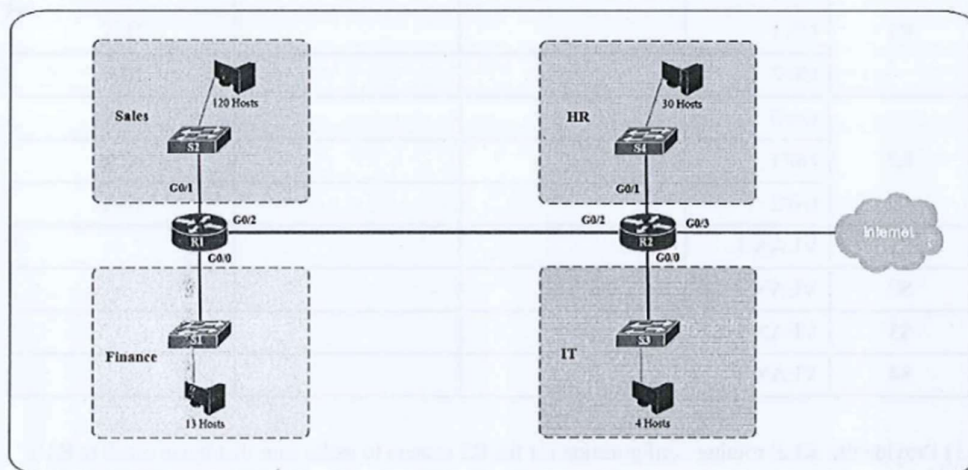
Answer Sheet for Section A (Multiple choice questions)

01	(A)	(B)	(C)	(D)
02	(A)	(B)	(C)	(D)
03	(A)	(B)	(C)	(D)
04	(A)	(B)	(C)	(D)
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27	(A)	(B)	(C)	(D)
28	(A)	(B)	(C)	(D)
29	(A)	(B)	(C)	(D)
30	(A)	(B)	(C)	(D)

Section B: Constructed-response questions (40 points)

A company has a network of $172.23.n.0/24$ that needs to provide IP addressing for its departments as follows (n is the order of students' name in the exam):



Design Requirements

- It is important to allocate IP addresses in a manner that minimizes IP address wastage.
- The subnets should be consecutive without any gaps between them.
- The first 10 IP addresses ($172.23.n.1 - 172.23.n.10$) should be reserved for the dedicated servers.
- The first usable IP address in each subnet is assigned for the router interface, while the last usable IP address is assigned for the switch.

(1) Complete the following table with the information of each created subnet (15 points)

Subnet Description	Network Address/CIDR	First Host Address	Last Host Address
HR			
IT			
Finance			
Sales			
R1-R2 link			

1

Device	Interface	IP Address	Subnet Mask (in Decimal)	Default Gateway
R1	G0/0			N/A
	G0/1			N/A
	G0/2			N/A
R2	G0/0			N/A
	G0/1			N/A
	G0/2			N/A
S1	VLAN 1			
S2	VLAN 1			
S3	VLAN 1			
S4	VLAN 1			

(3) Provide the detail routing configuration for the R2 routers to make sure that it can reach to R1's networks and reach to the Internet (10 points).

[illegible]

- - - - End - - - -

School of CIT: Approved

Sumit

Huynh Tan Phuoc

Lecturer



Phan Van Vinh



EXAM CODE: 02

Student's Full Name: Student ID:

Date: Room:

Section A: Multiple choice questions (60 points)

There are 30 multiple choice questions. Each question is worth **02 points**.

- Q1. What is the last host IP address in the subnet 10.20.24.192/27?
- A. 10.20.24.222 B. 10.20.24.206 C. 10.20.24.207 D. 10.20.24.254
- Q2. What is the network address for the IP address 192.168.10.155/27?
- A. 192.168.10.160 B. 192.168.10.32 C. 192.168.10.128 D. 192.168.10.64
- Q3. Which email protocol allows users to access and manage their email messages from multiple devices?
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- A. ip ssh version 2 B. aaa new-model
- C. ssh version 2 D. crypto key generate rsa
- Q13. Which DHCP message will a client send to accept an IP address that is offered by a DHCP server?
- A. DHCPACK B. DHCPOFFER C. DHCPDISCOVER D. DHCPREQUEST
- Q14. What is the purpose of an Intrusion Detection System (IDS)?
- A. To prevent unauthorized access to a network
- B. To monitor and detect suspicious activity on a network
- C. To encrypt and secure network communications
- D. To provide redundancy and fault tolerance in a network
- Q15. What is the purpose of a MAC address table in a Layer 2 switch?
- A. To determine the network segment of a device
- B. To identify the default gateway for a network
- C. To map IP addresses to MAC addresses
- D. To store information about connected devices' MAC addresses
- Q16. What type of attack involves an attacker intercepting communication between two parties?
- A. Phishing B. Man-in-the-middle C. Spoofing D. DoS

- Q17. What happens if a UDP packet is lost during transmission?
- A. The sender automatically retransmits the lost packet
 - B. The receiver sends a request to retransmit the lost packet
 - C. The sender and receiver negotiate a new connection and start the transmission again
 - D. The receiver discards the lost packet and continues with the next packet
- Q18. What is the purpose of TCP's three-way handshake?
- A. To terminate an established connection
 - B. To exchange acknowledgments between the sender and receiver
 - C. To ensure data integrity during transmission
 - D. To establish a connection between the client and server
- Q19. What is the primary defense against unauthorized remote access to network devices?
- A. Configuring a default gateway
 - B. Configuring a VTY password
 - C. Configuring an IP address
 - D. Configuring a console password
- Q20. Which prefix length is used for the subnet mask 255.255.255.224?
- A. /25
 - B. /27
 - C. /26
 - D. /24
- Q21. What is the subnet mask of the network 10.16.32.0/20?
- A. 255.255.252.0
 - B. 255.255.254.0
 - C. 255.255.240.0
 - D. 255.255.248.0
- Q22. What is the broadcast IP address in the subnet 10.20.24.0/21?
- A. 10.20.31.255
 - B. 10.20.24.255
 - C. 10.20.25.255
 - D. 10.20.27.255
- Q23. Which of the following protocols is used for receiving email?
- A. POP3
 - B. SMTP
 - C. IMAP
 - D. A and C
- Q24. What is the function of a router?
- A. It connects multiple IP networks.
 - B. It controls the flow of data via the use of Layer 2 addresses.
 - C. It determines the best path to send packets.
 - D. A and C
- Q25. Which of the following statements best describes ICMP protocol?
- A. It is a network layer protocol used for routing packets.
 - B. It is a transport layer protocol used for reliable data transmission.
 - C. It is an internet layer protocol used for error reporting and diagnostic purposes.
 - D. It is an application layer protocol used for remote network management.
- Q26. Which command is used to save the current running configuration on a Cisco switch?
- A. write
 - B. copy startup-config running-config
 - C. copy running-config startup-config
 - D. A and C
- Q27. Which of the following IPv4 addresses would not be used in a private network?
- A. 172.32.10.10
 - B. 172.20.1.12
 - C. 192.168.100.10
 - D. A and C
- Q28. What is the difference between authentication and authorization?
- A. Authentication verifies the identity of a user or device, while authorization determines what

actions they are allowed to perform

- B. Authentication secures wireless networks, while authorization manages network devices and configurations
 - C. Authentication encrypts data during transmission, while authorization determines network access permissions
 - D. Authentication prevents unauthorized access to a network, while authorization monitors network traffic for security threats
- Q29. When configuring a host, the gateway was not configured. How will this affect communication to and from this device?
- A. The host will not be able to communicate with devices by name.
 - B. No communication will be successful to or from the host.
 - C. Communication will be limited to hosts in the local network.
 - D. Host communication will not be affected.
- Q30. Which TCP mechanism is used to handle out-of-order packets?
- A. Three-way handshake
 - B. Congestion control
 - C. Selective Acknowledgment (SACK)
 - D. Sliding window

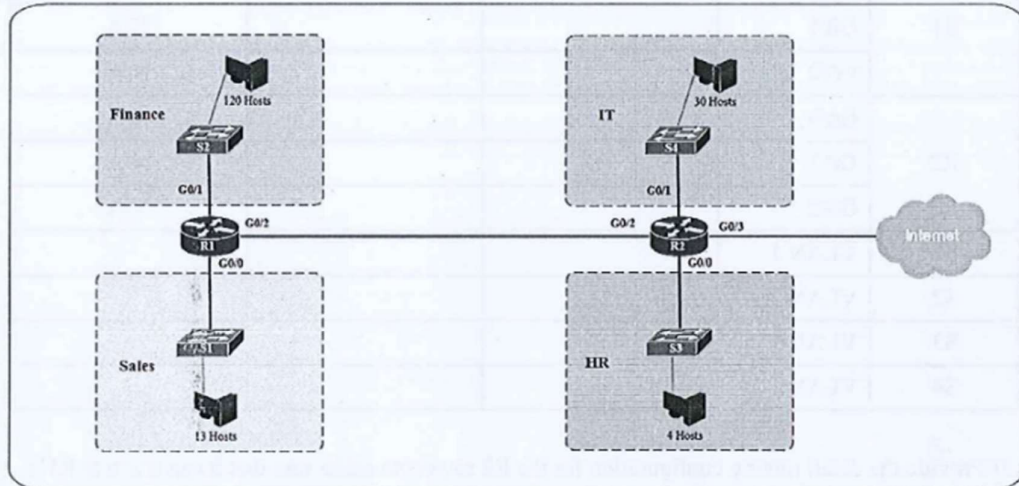
Answer Sheet for Section A (Multiple choice questions)

01	☐ A	☐ B	☐ C	☐ D
02	☐ A	☐ B	☐ C	☐ D
03	☐ A	☐ B	☐ C	☐ D
04	☐ A	☐ B	☐ C	☐ D
05	☐ A	☐ B	☐ C	☐ D
06	☐ A	☐ B	☐ C	☐ D
07	☐ A	☐ B	☐ C	☐ D
08	☐ A	☐ B	☐ C	☐ D
09	☐ A	☐ B	☐ C	☐ D
10	☐ A	☐ B	☐ C	☐ D
11	☐ A	☐ B	☐ C	☐ D
12	☐ A	☐ B	☐ C	☐ D
13	☐ A	☐ B	☐ C	☐ D
14	☐ A	☐ B	☐ C	☐ D
15	☐ A	☐ B	☐ C	☐ D

16	☐ A	☐ B	☐ C	☐ D
17	☐ A	☐ B	☐ C	☐ D
18	☐ A	☐ B	☐ C	☐ D
19	☐ A	☐ B	☐ C	☐ D
20	☐ A	☐ B	☐ C	☐ D
21	☐ A	☐ B	☐ C	☐ D
22	☐ A	☐ B	☐ C	☐ D
23	☐ A	☐ B	☐ C	☐ D
24	☐ A	☐ B	☐ C	☐ D
25	☐ A	☐ B	☐ C	☐ D
26	☐ A	☐ B	☐ C	☐ D
27	☐ A	☐ B	☐ C	☐ D
28	☐ A	☐ B	☐ C	☐ D
29	☐ A	☐ B	☐ C	☐ D
30	☐ A	☐ B	☐ C	☐ D

Section B: Constructed-response questions (40 points)

A company has a network of **10.20.n.0/24** that needs to provide IP addressing for its departments as follows (**n** is the order of students' name in the exam):



Design Requirements

- It is important to allocate IP addresses in a manner that minimizes IP address wastage.
- The subnets should be consecutive without any gaps between them.
- The first 10 IP addresses (*10.20.n.1 – 10.20.n.10*) should be reserved for the dedicated servers.
- The first usable IP address in each subnet is assigned for the router interface, while the last usable IP address is assigned for the switch.

(1) Complete the following table with the information of each created subnet (15 points)

Subnet Description	Network Address/CIDR	First Host Address	Last Host Address
HR			
IT			
Finance			
Sales			
R1-R2 link			

22

Device	Interface	IP Address	Subnet Mask (in Decimal)	Default Gateway
R1	G0/0			N/A
	G0/1			N/A
	G0/2			N/A
R2	G0/0			N/A
	G0/1			N/A
	G0/2			N/A
S1	VLAN 1			
S2	VLAN 1			
S3	VLAN 1			
S4	VLAN 1			

(3) Provide the detail routing configuration for the R2 routers to make sure that it can reach to R1's networks and reach to the Internet (10 points).

Year	Number of cases	Percentage of cases
1990	10	10.0
1991	15	15.0
1992	20	20.0
1993	25	25.0
1994	30	30.0
1995	35	35.0
1996	40	40.0
1997	45	45.0
1998	50	50.0
1999	55	55.0
2000	60	60.0
2001	65	65.0
2002	70	70.0
2003	75	75.0
2004	80	80.0
2005	85	85.0
2006	90	90.0
2007	95	95.0
2008	100	100.0
2009	105	105.0
2010	110	110.0
2011	115	115.0
2012	120	120.0
2013	125	125.0
2014	130	130.0
2015	135	135.0
2016	140	140.0
2017	145	145.0
2018	150	150.0
2019	155	155.0
2020	160	160.0
2021	165	165.0
2022	170	170.0
2023	175	175.0
2024	180	180.0
2025	185	185.0
2026	190	190.0
2027	195	195.0
2028	200	200.0
2029	205	205.0
2030	210	210.0
2031	215	215.0
2032	220	220.0
2033	225	225.0
2034	230	230.0
2035	235	235.0
2036	240	240.0
2037	245	245.0
2038	250	250.0
2039	255	255.0
2040	260	260.0
2041	265	265.0
2042	270	270.0
2043	275	275.0
2044	280	280.0
2045	285	285.0
2046	290	290.0
2047	295	295.0
2048	300	300.0
2049	305	305.0
2050	310	310.0
2051	315	315.0
2052	320	320.0
2053	325	325.0
2054	330	330.0
2055	335	335.0
2056	340	340.0
2057	345	345.0
2058	350	350.0
2059	355	355.0
2060	360	360.0
2061	365	365.0
2062	370	370.0
2063	375	375.0
2064	380	380.0
2065	385	385.0
2066	390	390.0
2067	395	395.0
2068	400	400.0
2069	405	405.0
2070	410	410.0
2071	415	415.0
2072	420	420.0
2073	425	425.0
2074	430	430.0
2075	435	435.0
2076	440	440.0
2077	445	445.0
2078	450	450.0
2079	455	455.0
2080	460	460.0
2081	465	465.0
2082	470	470.0
2083	475	475.0
2084	480	480.0
2085	485	485.0
2086	490	490.0
2087	495	495.0
2088	500	500.0
2089	505	505.0
2090	510	510.0
2091	515	515.0
2092	520	520.0
2093	525	525.0
2094	530	530.0
2095	535	535.0
2096	540	540.0
2097	545	545.0
2098	550	550.0
2099	555	555.0
2100		

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School of CIT: Approved

flannery

Huynh Tan Phuoc

Lecturer

Lecturer

Phan Van Vinh



EXAM CODE: 01

Student's Full Name: Student ID:

Date: Room:

Section A: Multiple choice questions (60 points)

There are 30 multiple choice questions. Each question is worth 02 points.

- Q1. Why is the DHCPREQUEST message sent as a broadcast during the DHCP process?
- A. For hosts on other subnets to receive the information
 - B. To notify other DHCP servers on the subnet that the IP address was leased
 - C. For routers to fill their routing tables with this new information
 - D. To notify other hosts not to request the same IP address
- Q2. Which command is used to save the current running configuration on a Cisco switch?
- A. write
 - B. copy startup-config running-config
 - C. copy running-config startup-config
 - D. A and C
- Q3. Which of the following statements about ARP spoofing is true?
- A. It is used to redirect network traffic to a different device
 - B. It is used to hide the MAC address of a device on a network
 - C. It is used to intercept network traffic between two devices
 - D. It is used to change the IP address of a device on a network
- Q4. Which command is used to view the current configuration of a Cisco router?
- A. enable
 - B. configure terminal
 - C. show running-config
 - D. show startup-config
- Q5. Which of the following is a common type of network attack?
- A. Denial of Service (DoS)
 - B. Social engineering
 - C. Phishing
 - D. All of the above
- Q6. Which of the following protocols is used for receiving email?
- A. POP3
 - B. SMTP
 - C. IMAP
 - D. A and C
- Q7. Which DHCP message will a client send to accept an IP address that is offered by a DHCP server?
- A. DHCPDISCOVER
 - B. DHCPREQUEST
 - C. DHCPACK
 - D. DHCPOFFER
- Q8. Which utility is used to identify network path between hosts?
- A. Ping
 - B. Traceroute
 - C. Tracert
 - D. B and C

- Q9. What is the correct order of data encapsulation?
- A. Data > frame > packet > segment > bit
 - B. Bit > frame > segment > packet > data
 - C. Bit > frame > packet > segment > data
 - D. Data > segment > packet > frame > bit
- Q10. Which of the following is a feature of the Transmission Control Protocol (TCP)?
- A. Connectionless protocol
 - B. No flow control
 - C. Reliable data delivery
 - D. A and C
- Q11. Which command is used to configure the default gateway on a Cisco switch?
- A. switchport trunk allowed vlan <vlan-list>
 - B. ip default-gateway <address>
 - C. interface vlan <vlan-id>
 - D. ip route <network> <mask> <gateway>
- Q12. A network technician borrows 5 bits from the 10.0.0.0/16 network to create subnets. How many host IP addresses are there in each created subnet?
- A. 254
 - B. 2048
 - C. 30
 - D. 1022
- Q13. Which of the following IPv4 addresses would not be used in a private network?
- A. 172.10.10.10
 - B. 172.20.1.12
 - C. 192.168.100.10
 - D. A and C
- Q14. Which of the following statements best describes ICMP protocol?
- A. It is an internet layer protocol used for error reporting and diagnostic purposes.
 - B. It is a transport layer protocol used for reliable data transmission.
 - C. It is an application layer protocol used for remote network management.
 - D. It is a network layer protocol used for routing packets.
- Q15. What is the network address for the IP address 192.168.10.55/27?
- A. 192.168.10.0
 - B. 192.168.10.32
 - C. 192.168.10.48
 - D. 192.168.10.64
- Q16. What type of message does a device send when it needs to resolve an IP address to a MAC address?
- A. ARP request message
 - B. ICMP message
 - C. UDP message
 - D. ARP response message
- Q17. What is the broadcast IP address in the subnet 198.10.100.64/27?
- A. 198.10.100.63
 - B. 198.10.100.96
 - C. 198.10.100.95
 - D. 198.10.100.127
- Q18. Which the protocol does email users and servers use to process email?
- A. SMTP
 - B. IMAP
 - C. POP3
 - D. All of the above
- Q19. When configuring a host, the gateway was not configured. How will this affect communication to and from this device?
- A. Host communication will not be affected.
 - B. The host will not be able to communicate with devices by name.
 - C. No communication will be successful to or from the host.
 - D. Communication will be limited to hosts in the local network.

- Q20. What is the primary defense against unauthorized remote access to network devices?
- A. Configuring a VTY password
 - B. Configuring a console password
 - C. Configuring an IP address
 - D. Configuring a default gateway
- Q21. Which prefix length is used for the subnet mask 255.255.255.192?
- A. /24
 - B. /25
 - C. /26
 - D. /27
- Q22. Which devices on a network will receive an ARP request?
- A. Only the device that has the IPv4 address that the request is looking for
 - B. Only devices in the same collision domain
 - C. All devices in the Layer 2 broadcast domain
 - D. All devices in the entire networks
- Q23. What is a characteristic of UDP?
- A. Reliable delivery
 - B. Expectational acknowledgements
 - C. Connectionless
 - D. Flow control
- Q24. A new department has been established with 31 hosts that require addresses. Currently the company uses the 10.10.113.0/24 address space. How many bits must the network administrator borrow to provide addresses for this subnet?
- A. 2
 - B. 3
 - C. 4
 - D. 5
- Q25. What is the function of a Layer 2 switch?
- A. Duplicates the electrical signal of each frame to every port
 - B. Forwards data based on logical addressing
 - C. Determines which interface is used to forward a frame based on the destination MAC address
 - D. Learns the port assigned to a host by examining the destination MAC address
- Q26. What is the subnet mask of the network 10.16.0.0/23?
- A. 255.255.248.0
 - B. 255.255.254.0
 - C. 255.255.252.0
 - D. 255.255.224.0
- Q27. What is one difference between using Telnet or SSH to connect to a network device for management purposes?
- A. Telnet uses UDP as the transport protocol, whereas SSH uses TCP
 - B. Telnet does not provide authentication, whereas SSH does.
 - C. Telnet sends a username and password in plain text, whereas SSH encrypts the username and password.
 - D. Telnet supports a host GUI, whereas SSH only supports a host CLI.
- Q28. What is the function of a router?
- A. It connects multiple IP networks.
 - B. It controls the flow of data via the use of Layer 2 addresses.
 - C. It determines the best path to send packets.
 - D. A and C
- Q29. What type of attack involves an attacker intercepting communication between two parties?
- A. Spoofing
 - B. Man-in-the-middle
 - C. Denial of service
 - D. Phishing
-

Q30. Which command is used to configure SSH access on a Cisco router?

- A. crypto key generate rsa B. ip ssh version 2
C. aaa new-model D. ssh version 2

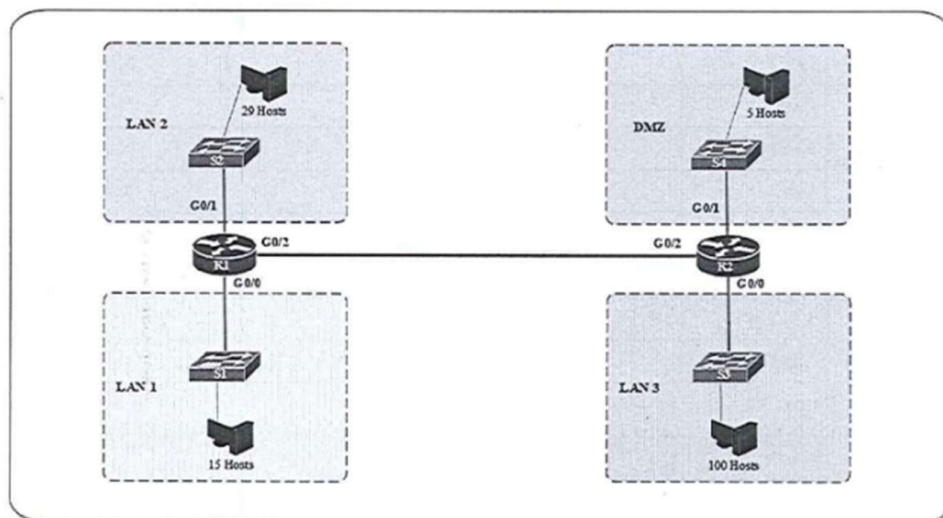
Answer Sheet for Section A (Multiple choice questions)

01	(A)	(B)	(C)	(D)
02	(A)	(B)	(C)	(D)
03	(A)	(B)	(C)	(D)
04	(A)	(B)	(C)	(D)
05	(A)	(B)	(C)	(D)
06	(A)	(B)	(C)	(D)
07	(A)	(B)	(C)	(D)
08	(A)	(B)	(C)	(D)
09	(A)	(B)	(C)	(D)
10	(A)	(B)	(C)	(D)
11	(A)	(B)	(C)	(D)
12	(A)	(B)	(C)	(D)
13	(A)	(B)	(C)	(D)
14	(A)	(B)	(C)	(D)
15	(A)	(B)	(C)	(D)

16	(A)	(B)	(C)	(D)
17	(A)	(B)	(C)	(D)
18	(A)	(B)	(C)	(D)
19	(A)	(B)	(C)	(D)
20	(A)	(B)	(C)	(D)
21	(A)	(B)	(C)	(D)
22	(A)	(B)	(C)	(D)
23	(A)	(B)	(C)	(D)
24	(A)	(B)	(C)	(D)
25	(A)	(B)	(C)	(D)
26	(A)	(B)	(C)	(D)
27	(A)	(B)	(C)	(D)
28	(A)	(B)	(C)	(D)
29	(A)	(B)	(C)	(D)
30	(A)	(B)	(C)	(D)

Section B: Constructed-response questions (40 points)

A company has a network of 192.168.n.0/24 that needs to provide IP addressing for LANs as follows (n is the order of students' name in the exam):



Design Requirements

- Provide the most efficient subnet possible for the point-to-point link between the routers.
- The first range of IP addresses should be reserved for the hosts in the DMZ area (this is a special LAN for dedicated servers).

(1) Complete the following table with the information of each created subnet (15 points)

Subnet Description	Network Address/CIDR	First Host Address	Last Host Address
LAN 1			
LAN 2			
LAN 3			
DMZ			
R1-R2 link			



EXAM CODE: 02

Student's Full Name: Student ID:

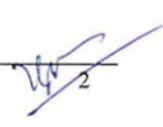
Date: Room:

Section A: Multiple choice questions (60 points)

There are 30 multiple choice questions. Each question is worth 02 points.

- Q1. What is the function of a router?
- A. It connects multiple IP networks. B. It controls the flow of data via the use of Layer 2 addresses.
- C. It determines the best path to send packets. D. A and C
- Q2. What is the network address for the IP address 192.168.10.55/27?
- A. 192.168.10.0 B. 192.168.10.32 C. 192.168.10.48 D. 192.168.10.64
- Q3. Which of the following statements best describes ICMP protocol?
- A. It is a transport layer protocol used for reliable data transmission.
- B. It is an internet layer protocol used for error reporting and diagnostic purposes.
- C. It is a network layer protocol used for routing packets.
- D. It is an application layer protocol used for remote network management.
- Q4. Which of the following is a common type of network attack?
- A. Denial of Service (DoS) B. Social engineering
- C. Phishing D. All of the above
- Q5. Which devices on a network will receive an ARP request?
- A. Only the device that has the IPv4 address that the request is looking for
- B. Only devices in the same collision domain
- C. All devices in the Layer 2 broadcast domain
- D. All devices in the entire networks
- Q6. Which command is used to view the current configuration of a Cisco router?
- A. show running-config B. configure terminal C. enable D. show startup-config
- Q7. Which command is used to configure SSH access on a Cisco router?
- A. ip ssh version 2 B. ssh version 2
- C. aaa new-model D. crypto key generate rsa

- Q8. What is the broadcast IP address in the subnet 198.10.100.64/27?
- A. 198.10.100.63 B. 198.10.100.127 C. 198.10.100.95 D. 198.10.100.96
- Q9. Which utility is used to identify network path between hosts?
- A. Ping B. Traceroute C. Tracert D. B and C
- Q10. What is the correct order of data encapsulation?
- A. Data > frame > packet > segment > bit B. Data > segment > packet > frame > bit
C. Bit > frame > segment > packet > data D. Bit > frame > packet > segment > data
- Q11. What is the primary defense against unauthorized remote access to network devices?
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C. Configuring a VTY password D. Configuring a default gateway
- Q12. Which DHCP message will a client send to accept an IP address that is offered by a DHCP server?
- A. DHCPREQUEST B. DHCPACK C. DHCPDISCOVER D. DHCPOFFER
- Q13. Which the protocol does email users and servers use to process email?
- A. SMTP B. IMAP C. POP3 D. All of the above
- Q14. Why is the DHCPREQUEST message sent as a broadcast during the DHCP process?
- A. To notify other DHCP servers on the subnet that the IP address was leased
B. For routers to fill their routing tables with this new information
C. For hosts on other subnets to receive the information
D. To notify other hosts not to request the same IP address
- Q15. Which command is used to save the current running configuration on a Cisco switch?
- A. write B. copy startup-config running-config
C. copy running-config startup-config D. A and C
- Q16. What is the subnet mask of the network 10.16.0.0/23?
- A. 255.255.248.0 B. 255.255.252.0 C. 255.255.224.0 D. 255.255.254.0
- Q17. Which of the following IPv4 addresses would not be used in a private network?
- A. 172.10.10.10 B. 172.20.1.12 C. 192.168.100.10 D. A and C
- Q18. What is the function of a Layer 2 switch?
- A. Learns the port assigned to a host by examining the destination MAC address
B. Forwards data based on logical addressing
C. Duplicates the electrical signal of each frame to every port
D. Determines which interface is used to forward a frame based on the destination MAC address
- Q19. Which prefix length is used for the subnet mask 255.255.255.192?
- A. /24 B. /25 C. /26 D. /27



- Q20. Which command is used to configure the default gateway on a Cisco switch?
- A. switchport trunk allowed vlan <vlan-list>
 - B. ip default-gateway <address>
 - C. interface vlan <vlan-id>
 - D. ip route <network> <mask> <gateway>
- Q21. What is one difference between using Telnet or SSH to connect to a network device for management purposes?
- A. Telnet uses UDP as the transport protocol, whereas SSH uses TCP
 - B. Telnet supports a host GUI, whereas SSH only supports a host CLI.
 - C. Telnet does not provide authentication, whereas SSH does.
 - D. Telnet sends a username and password in plain text, whereas SSH encrypts the username and password.
- Q22. What type of attack involves an attacker intercepting communication between two parties?
- A. Spoofing
 - B. Phishing
 - C. Denial of service
 - D. Man-in-the-middle
- Q23. Which of the following protocols is used for receiving email?
- A. POP3
 - B. SMTP
 - C. IMAP
 - D. A and C
- Q24. Which of the following is a feature of the Transmission Control Protocol (TCP)?
- A. Connectionless protocol
 - B. No flow control
 - C. Reliable data delivery
 - D. A and C
- Q25. A network technician borrows 5 bits from the 10.0.0.0/16 network to create subnets. How many host IP addresses are there in each created subnet?
- A. 30
 - B. 2048
 - C. 1022
 - D. 254
- Q26. Which of the following statements about ARP spoofing is true?
- A. It is used to redirect network traffic to a different device
 - B. It is used to hide the MAC address of a device on a network
 - C. It is used to intercept network traffic between two devices
 - D. It is used to change the IP address of a device on a network
- Q27. When configuring a host, the gateway was not configured. How will this affect communication to and from this device?
- A. Host communication will not be affected.
 - B. The host will not be able to communicate with devices by name.
 - C. Communication will be limited to hosts in the local network.
 - D. No communication will be successful to or from the host.
- Q28. A new department has been established with 31 hosts that require addresses. Currently the company uses the 10.10.113.0/24 address space. How many bits must the network administrator borrow to provide addresses for this subnet?
- A. 2
 - B. 3
 - C. 4
 - D. 5

Q29. What type of message does a device send when it needs to resolve an IP address to a MAC address?

- A. UDP message B. ARP request message C. ICMP message D. ARP response message

Q30. What is a characteristic of UDP?

- A. Reliable delivery B. Expectational acknowledgements
C. Connectionless D. Flow control

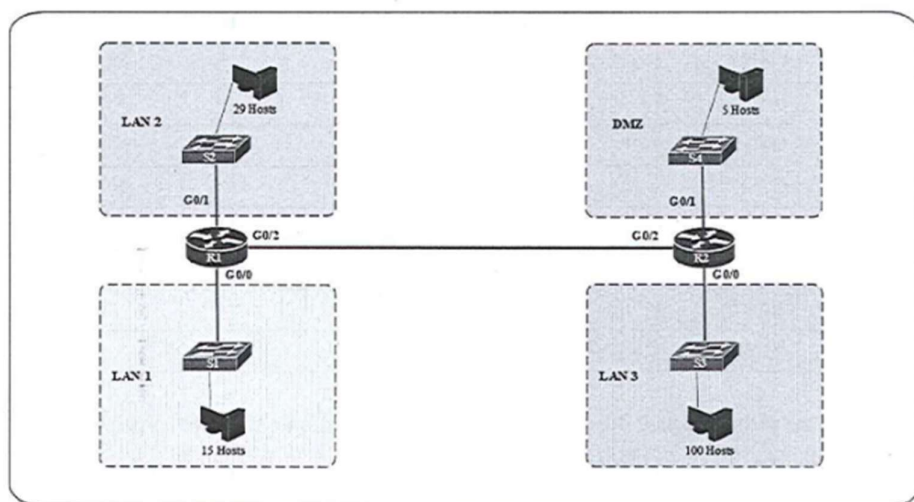
Answer Sheet for Section A (Multiple choice questions)

01	(A)	(B)	(C)	(D)
02	(A)	(B)	(C)	(D)
03	(A)	(B)	(C)	(D)
04	(A)	(B)	(C)	(D)
05	(A)	(B)	(C)	(D)
06	(A)	(B)	(C)	(D)
07	(A)	(B)	(C)	(D)
08	(A)	(B)	(C)	(D)
09	(A)	(B)	(C)	(D)
10	(A)	(B)	(C)	(D)
11	(A)	(B)	(C)	(D)
12	(A)	(B)	(C)	(D)
13	(A)	(B)	(C)	(D)
14	(A)	(B)	(C)	(D)
15	(A)	(B)	(C)	(D)

16	(A)	(B)	(C)	(D)
17	(A)	(B)	(C)	(D)
18	(A)	(B)	(C)	(D)
19	(A)	(B)	(C)	(D)
20	(A)	(B)	(C)	(D)
21	(A)	(B)	(C)	(D)
22	(A)	(B)	(C)	(D)
23	(A)	(B)	(C)	(D)
24	(A)	(B)	(C)	(D)
25	(A)	(B)	(C)	(D)
26	(A)	(B)	(C)	(D)
27	(A)	(B)	(C)	(D)
28	(A)	(B)	(C)	(D)
29	(A)	(B)	(C)	(D)
30	(A)	(B)	(C)	(D)

Section B: Constructed-response questions (40 points)

A company has a network of 10.10.n.0/24 that needs to provide IP addressing for LANs as follows (n is the order of students' name in the exam):



Design Requirements

- Provide the most efficient subnet possible for the point-to-point link between the routers.
- The first range of IP addresses should be reserved for the hosts in the DMZ area (this is a special LAN for dedicated servers).

(1) Complete the following table with the information of each created subnet (15 points)

Subnet Description	Network Address/CIDR	First Host Address	Last Host Address
LAN 1			
LAN 2			
LAN 3			
DMZ			
R1-R2 link			

